Supplementary Materials: Privacy, Utility, and Explainability Trade-offs in Privacy-Preserving Federated Recommender Systems

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Overview

This document provides supplementary tables and technical specifications supporting the manuscript submitted to *Data Technologies and Applications* (Emerald Publishing). The contents include:

- **Table S1:** Comprehensive literature positioning comparing selected federated recommendation, privacy, and explainability studies.
- **Table S2:** Complete system hyperparameter specifications across model architecture, federated learning, differential privacy, homomorphic encryption, and post-hoc explainers.
- **Table S3:** Cross-dataset validation comparing empirical ranking, latency, and explanation sensitivity metrics between the proprietary enterprise e-commerce benchmark (EARS) and the public movie benchmark (MovieLens-100k).

Table 1: Comparison of selected studies in federated recommendation, privacy, and explainability.

Study	**FL Setting**	**HE Scheme**	**DP Analysis**	**XAI Methods**	**Ranking Metrics**	**Trade-off Analysis**
Tan et al. (2020)	FedAvg	None	None	None	Top-K ranking	Not reported
Minto et al. (2021)	Secure aggregation	Paillier/CKKS	None	None	NDCG, Recall	Communication–utility
Javeed et al. (2023)	FedAvg	None	Gaussian DP	None	MSE, Precision	Privacy–utility
Saha et al. (2024)	Survey	Conceptual	Conceptual	None	Not applicable	Qualitative review
Shin et al. (2023)	FedAvg	None	Latent DP	None	Hit Rate, NDCG	Privacy–utility
Gaspar et al. (2025)	FedAvg	None	None	SHAP, LIME	Accuracy, RMSE	XAI–accuracy
Gaspar et al. (2026)	Review	Partial	Partial	None	Not applicable	Literature synthesis
This study	**FedAvg**	**TenSEAL CKKS**	**Client RDP** ($\sigma \in \{0.5, 1.0, 2.0\}$)	**KernelSHAP, LIME**	**P, R, NDCG, MAP, HR**	**Four-axis comparison**

Source: Authors' own work.

Table 2: System hyperparameter specifications across experimental modules.

Module	Hyperparameter	Value or Specification
Model Architecture	Base recommender	`CoreHybridRecSys` multi-branch neural network
	Trainable parameters	586,700 weights (2, 346.8 KB, float32)
	Feature dimensionality	43 tabular features across user state, collaborative, and context branches
	Loss formulation	Multi-task: BCE (1.0) + Margin Ranking (0.2) + Confidence (0.1)
Federated Learning	Total clients	$N = 20$ client partitions
	Sampled clients/round	$m = 5$ clients (25% participation rate, $q = 0.25$)
	Communication rounds	$T = 15$ global rounds
	Local training	$E = 1$ epoch, Adam optimizer, $\eta = 0.001$, batch size 64
Differential Privacy	Privacy unit	Client partition level
	Adjacency relation	Client replacement adjacency ($\mathcal{D} \sim \mathcal{D}'$)
	Clipping norm	$C = 1.0$ (L_2 client update clipping bound)
	Noise multipliers	$\sigma \in \{0.5, 1.0, 2.0\}$ (nominal presets DP-1, DP-2, DP-3)
	Target δ & Accountant	$\delta = 10^{-5}$; Subsampled Gaussian RDP ($\epsilon \in \{11.17, 5.12, 2.44\}$, provisional)
Homomorphic Encryption	Cryptographic scheme	TenSEAL CKKS (Microsoft SEAL 3.7.2 backend)
	Polynomial degree	$N_{\text{poly}} = 8{,}192$ (SIMD slot capacity: 4,096 complex slots)
	Modulus bit sizes	$[60, 40, 40, 60]$ (200 total modulus bits; 128-bit classical security)
	Scale bit width	2^{40} precision scale
Explainability (XAI)	Evaluation instances	$M_{\text{eval}} = 30$ instances across holdout test users
	KernelSHAP budget	$K_{\text{bg}} = 20$ background instances, 50 coalition samples/instance (1,500 model evals)
	LIME budget	500 normal perturbation samples/instance, local ridge surrogate (15,000 model evals)
	Masking parameter	Top-3 features by descending absolute SHAP attribution
Datasets	EARS benchmark	5,157 interaction samples partitioned across 20 training clients (mean: 257.8)
	MovieLens-100k	8,255 filtered interactions partitioned across 20 clients ($\alpha = 0.5$; 50 test users)

Source: Authors' own work. For EARS, the full constructed pool contains 14,024 *samples across 50 users, of which* 5,157 *are allocated across the 20 federated training clients.*

Table 3: Cross-dataset validation: comparison between proprietary enterprise e-commerce (EARS) and public entertainment benchmark (MovieLens-100k) across configurations (mean ± standard deviation across seeds $\{42, 43, 44\}$).

Config	NDCG@10		Recall@10		Train Latency (s)		Top-3 Masking Sensitivity Δ	
	EARS	**ML-100k**	**EARS**	**ML-100k**	**EARS**	**ML-100k**	**EARS**	**ML-100k**
E0	0.526 ± 0.002	0.913 ± 0.032	0.384 ± 0.017	0.370 ± 0.000	27.3 ± 1.8	17.6 ± 0.9	3.76 ± 0.24	3.10 ± 0.40
E1	0.563 ± 0.030	0.888 ± 0.035	0.406 ± 0.011	0.370 ± 0.000	25.3 ± 1.7	18.8 ± 0.3	4.00 ± 0.68	2.39 ± 1.13
E2_DP1	0.522 ± 0.106	0.740 ± 0.033	0.236 ± 0.160	0.367 ± 0.005	25.3 ± 1.2	19.3 ± 0.3	0.80 ± 0.37	0.59 ± 0.50
E2_DP2	0.430 ± 0.082	0.402 ± 0.314	0.161 ± 0.075	0.151 ± 0.190	25.5 ± 0.5	19.9 ± 0.7	2.97 ± 0.77	2.74 ± 3.70
E2_DP3	0.402 ± 0.140	0.273 ± 0.115	0.108 ± 0.130	0.077 ± 0.067	26.5 ± 2.0	18.8 ± 0.5	11.70 ± 5.66	4.07 ± 2.65
E3	0.565 ± 0.030	0.894 ± 0.039	0.407 ± 0.011	0.370 ± 0.000	49.7 ± 0.7	44.0 ± 0.6	3.99 ± 0.66	2.38 ± 1.32
E4	0.434 ± 0.038	0.375 ± 0.170	0.214 ± 0.068	0.146 ± 0.110	51.2 ± 2.0	45.5 ± 0.5	0.58 ± 0.36	0.96 ± 0.27

Source: Authors' own work.